LiDAR2CAD: Joint Structural Reconstruction From LiDAR and Design CAD for Underground Parking Garages

***Abstract*— Industrial digital twins and intelligent facility management require traceable spatial information that reconciles field measurements with engineering designs. In underground parking garages, this information must integrate structural and functional elements despite design discrepancies and incomplete observations. Design drawings may retain outdated structures, whereas vehicle occlusion and uneven laser-scanning coverage fragment observed walls and obscure connections. Simply combining these sources can therefore preserve incorrect geometry or close open passages. This paper presents LiDAR2CAD, which couples joint structural reconstruction with facility-layer mapping to produce multilayer semantic computer-aided design maps. Unique voxels associate observations across heights. Design directions constrain wall fitting, reliable observations update supported design intervals, and observed anchors, design continuity, and conflict checks guide missing-structure recovery. Parking spaces, lane markings, and directional arrows are mapped from design layers. The output distinguishes observed geometry, conditional inference, and unverified design retention. Experiments on four ICPARK evaluation regions achieve wall and connection F1 scores of 86.44% and 92.31%, respectively. Controlled comparisons support the benefit of bidirectional correction, and three additional facilities show improvements over design-only reconstruction. These results support traceable structural recovery, while large design discrepancies and difficult regions remain challenging.**

***Index Terms*— Computer-aided design, digital twins, light detection and ranging, semantic mapping, structural reconstruction, underground parking garages.**

I. Introduction

A. Background

Industrial digital twins and intelligent facility management require consistency between physical environments and digital models through verifiable, updatable spatial information [1], [2]. In underground garages, automated valet parking and multivehicle planning depend on parking layouts and passage connectivity [3], [4]. Semantic maps must integrate structural boundaries—walls, columns, and their connections—with functional elements—parking spaces, lane markings, and directional arrows—in common coordinates. Missing elements and noisy connections can disrupt road-network extraction [5].

Such maps must reconcile drawings with current conditions. Design or as-built computer-aided design (CAD) drawings record intended or delivered configurations, which require field verification and updating [6]. Mobile light detection and ranging (LiDAR) captures large-scale three-dimensional observations, but not editable structural objects and connections.



(a) Design CAD (b) LiDAR-supported reconstruction

Fig. 1. Matched full-scene ICPARK views. Red ellipses mark corresponding regions with design–field discrepancies, including walls absent from the design and recovered from LiDAR. Extent and scale are identical; other retained design geometry is not thereby field-verified.

Semantic classification identifies structural categories [7], [8]; reconstruction must additionally associate same-object observations, estimate geometry, and determine connections.

Design drawings can also omit walls that are observed on site. Fig. 1 compares matched full-scene ICPARK views: the circled regions contain wall sections that are absent from the design but appear in the LiDAR-supported reconstruction. Design must therefore act as a revisable prior, rather than an exhaustive inventory of existing structures.

Repetitive geometry, restricted coverage, and vehicle interference complicate reconstruction [9]. Regular columns and repeated walls create association ambiguities. Lane-following platforms provide uneven lateral coverage, while parked vehicles occlude structures and generate confusing returns. A wall may thus appear continuous above vehicles but fragmented or absent below. Overlapping vehicle and structural heights prevent a single crop from both preserving structure and suppressing interference. Independent slice vectorization produces misaligned fragments, while merged planar projections discard height provenance and correspondences.

Design CAD complements observations with layout, entity identity, and continuity, raising three questions: 1) *design discrepancy correction*: how can reliable observations correct local drafting errors, construction deviations, or later modifications? 2) *missing-structure recovery*: how can design restore geometry and connections affected by occlusion, reconstruction errors, and object fragmentation? 3) *completion reliability*: without direct observations, how can supported inference be distinguished from unverified design retention to avoid outdated structures and closed passages? We investigate joint reconstruction with mutual design–observation constraints.

B. Related Work

Projection-based vectorization and geometric primitives. Projected density or edge maps support classical line and contour extraction. Hough voting [10], progressive probabilistic Hough sampling [11], and random sample consensus (RANSAC) [12] estimate geometric primitives. Douglas–Peucker simplification reduces contour vertices [13], and Zhang–Suen thinning extracts binary skeletons [14]. The line segment detector (LSD), fast line detector (FLD), and EDLines directly detect planar segments [15]–[17]. We evaluate these seven baselines as P. Hough, DP-Contour, RANSAC, Morph-Skel, LSD, FLD, and EDLines. DP-Contour and Morph-Skel combine contour extraction or morphology with the cited simplification or thinning steps. Their output does not explicitly preserve cross-height observation ownership or establish structural identities and connections.

Point-cloud projection and structured floorplans. Top-down density, occupancy, and features convert scans into planar structures. FloorNet combines point-cloud, image, and top-down features [18]; Floor-SP recovers room contours through sequential shortest paths [19]; and MonteFloor optimizes polygons through Monte Carlo tree search [20]. HEAT, RoomFormer, PolyRoom, and FRI-Net use edge relationships, queries, polygon prediction, or implicit representations [21]–[24]. However, open lanes, long walls, and isolated garage columns do not uniformly form closed rooms. Planar representations without height provenance also struggle to distinguish structural discontinuities from occlusion gaps.

Semantic understanding and cross-height association. Semantic features distinguish structures from vehicles and equipment. 3DGraphSeg and the field-aware transformer (FAT) segment industrial point clouds using graphs and adaptive receptive fields [7], [8]; Point Transformer V3 and Sonata provide large-scale point representations [25], [26]. Yet neighboring wall-labeled fragments need not share a physical surface. Cross-height association requires position, orientation, and vertical persistence, retaining original-return links for consistent updates.

Incomplete observations and structural topology. Sun et al. [27] complete small industrial objects using a density-sensitive transformer, addressing geometry and point distribution. Structural reconstruction also requires component relationships: Kim and Kim [28] use connectivity to recover incomplete structural geometry. Orientation-space minimal paths with curvature penalties exploit tangents and continuity [29] and can generate gap-bridging candidates. However, proximity and continuity cannot establish connections among parallel walls, column grids, and open lanes. Fragmentation also changes evidence ownership beside a candidate. Recovery therefore requires host geometry, observation support, and spatial conflicts, not geometric continuity alone.

Design priors and field reconstruction. CAD and building information models supply geometry, spatial organization, and component identities [6]. Zhou et al. [2] maintain site-object states by matching camera observations to building information models through coordinate transformations. ParkingTwin uses CAD priors for three-dimensional parking-facility skeletons [9]. Fixed priors constrain missing areas but cannot incorporate observation-driven corrections; postreconstruction matching cannot constrain earlier fitting. Correspondences should instead participate in geometric updates, while completion distinguishes observation-constrained inference from unverified design.

C. Contributions

LiDAR2CAD combines joint wall–column reconstruction with parking-space, lane-marking, and directional-arrow mapping in a common semantic CAD map. Its methodological contributions and quantitative evaluation concern the structural layer:

1) **Object-level representation of complementary observations across heights.** Unique three-dimensional voxels support fragment association using position, orientation, and vertical persistence, providing deduplicated evidence for fitting and candidate verification while retaining observation ownership.
2) **Bidirectional geometric correction between design and LiDAR.** Alternating updates use design directions to constrain observed walls and reliable observations to update effective design trajectories. Correspondences are reestimated, and residual and displacement bounds constrain adjustments.
3) **Evidence-conditioned reconstruction separating completion from field verification.** Observed anchors, design continuity, and conflicts govern missing intervals. Conditional inference and unverified design retention are distinguished, with structural status, geometric provenance, and visibility recorded independently.

The remainder of this paper is organized as follows. Section II defines the reconstruction inputs, object identities, and interval-level status, provenance, and visibility attributes. Section III describes cross-height structural initialization, bidirectional design–LiDAR correction, evidence-conditioned regularization, and multilayer CAD generation. Section IV presents the datasets, evaluation metrics, and baselines, followed by geometric and topological results, controlled comparisons, ablations, robustness and failure analysis, efficiency, and additional-facility tests. Section V summarizes the findings, limitations, and future work.

II. Problem Formulation and Structural Representation

A. Inputs, Outputs, and Identities

Given a gravity-aligned point cloud $\mathcal{P} = \{(\mathbf{p}_i, q_i)\}$ and same-floor design drawing $\mathcal{D}_0 = (\mathcal{W}_d, \mathcal{C}_d, \mathcal{H}_d)$, q_i is a semantic wall probability and the design sets contain walls, columns, and openings. Registered rays or explicit scan coverage, when available, form evidence $\mathcal{E}_s$; otherwise, $\mathcal{E}_s = \varnothing$. A fixed global transformation maps design into site coordinates. The recovered structural layer is

$$\mathcal{M} = (\mathcal{O}, \mathcal{N}, \mathcal{R}, \mathcal{A}), \tag{1}$$

where $\mathcal{O}$ contains wall trajectories and column cross sections, $\mathcal{N}$ endpoints and junctions, $\mathcal{R}$ accepted structural connections, and $\mathcal{A}$ interval attributes and provenance. Observed, conditionally inferred, and retained design geometry constitute $\mathcal{M}$. Mapped parking spaces, lane markings, and arrows form facility layers; together they yield the semantic CAD map. Quantitative reconstruction evaluates $\mathcal{M}$, whereas mapping a facility layer does not confer field verification.

A wall face is a single-sided trajectory $\Gamma_o : [0, L_o] \to \mathbb{R}^2$; opposite faces remain separate. A column is a closed boundary ∂C_o. Site-object i_o, design-entity j_d, and reference k_g identifiers are distinct; site-to-design correspondence may be one-to-many, and fragments retain parent identities. Relation $r = (i_o, i_{o'}, \gamma_r)$ records an accepted connection and path. Geometric contact supports connectivity statistics but does not by itself create a relation in $\mathcal{R}$.

B. Status, Provenance, and Visibility

Each wall interval $I \subseteq [0, L_o]$ has independent attributes

$$\begin{aligned} \sigma(I) &\in \{\text{present}, \text{open}, \text{undecided}\}, \\ \rho(I) &\in \{\text{observed}, \text{inferred}, \text{design}\}, \\ \nu(I) &\in \{\text{seen}, \text{occluded}, \text{unscanned}, \text{unknown}\}. \end{aligned} \tag{2}$$

Here, σ denotes structural status, ρ geometric provenance, and ν sensor visibility. Directly supported walls may be $(\text{present}, \text{observed}, \text{seen})$; two-sided gap inference may be $(\text{present}, \text{inferred}, \text{unknown})$; retained design is $(\text{undecided}, \text{design}, \text{unscanned})$ or $(\text{undecided}, \text{design}, \text{unknown})$. Thus, retaining design does not verify on-site presence.

Evidence $\mathbf{e}(I) = (n_v, n_z, c_o, \mathcal{I}_o, \mathcal{J}_d, f_I)$ records unique voxels, independent height bins, host-compatible coverage, source voxels, design correspondences, and free-space evidence. Missing returns remove direct support but do not determine σ. Occluded or unscanned states require poses, rays, or explicit coverage; otherwise, $\nu = \text{unknown}$.

C. Complete Geometry and Evaluation

Final wall–column geometry is

$$\mathcal{G}_{\text{final}} = \mathcal{G}_{\text{obs}} \cup \mathcal{G}_{\text{inf}} \cup \mathcal{G}_{\text{prior}}. \tag{3}$$

The sets follow supported-interval replacement, deduplication, and conflict checks, rather than raw observation–design overlay. Common-region metrics evaluate complete geometry; provenance statistics separate source contributions. Walls and column boundaries are scored separately, connections by $\mathcal{R}$, and open passages by crossings of complete geometry, measuring placement, relations, and spatial conflicts, respectively.

III. Bidirectional Design–LiDAR Correction and Evidence-Conditioned Regularization

Fig. 2 summarizes initialization, correspondence, bidirectional correction, and regularization.

A. Cross-Height Structural Initialization

The point cloud is partitioned into unique voxels $\mathcal{V}$ of horizontal resolution g and height-bin width Δz:

$$v_i = \left(\left\lfloor \frac{x_i - x_0}{g} \right\rfloor, \left\lfloor \frac{y_i - y_0}{g} \right\rfloor, \left\lfloor \frac{z_i - z_f}{\Delta z} \right\rfloor \right), \tag{4}$$

where z_f is ground height. Thin and thick windows use width/step 0.50/0.30 and 1.00/0.50 m, respectively, sweeping from floor to ceiling with a ceiling-aligned tail window. Shorter spans use one clipped window. Windows reference voxel indices to preserve low edges and structures spanning occlusions without duplicating measurements.

Object support unions fragment voxel indices: repeated membership counts once, and height bins use original vertical coordinates.

A wall fragment $f_i = (\Gamma_i, \mathbf{t}_i, [a_i, b_i], \mathcal{I}_i)$ contains its planar trajectory, unoriented tangent, height span, and source voxels. Group voxels jointly fit walls; projected extents define endpoints. Semantic filtering rejects low-confidence candidates without moving voxels.

Candidate pairs from different windows require orientation difference $\leq 10^\circ$, symmetric lateral deviation ≤ 0.035 m, vertical separation ≤ 1.00 m, and projected overlap $\geq \min(0.15\,\text{m}, 0.30 \min(\ell_i, \ell_j))$ for fragment lengths ℓ_i, ℓ_j. Edges follow increasing line distance with stable-ID ties. Every cross-group pair must pass direction and lateral tests before merging, preventing transitive drift across parallel walls. Deduplication prevents inflated support, not semantic errors.

Compact, scale-stable multiheight structures yield rectangular or circular columns. Reliable boundaries constrain wall extension; candidates with insufficient visible faces remain uncertain instead of using their centroid as a column center. Missing initial wall fragments are sought along site trajectories, fused candidates, and registered design corridors, then verified by cross-height support, longitudinal occupancy, and residuals. Design defines only the search corridor; accepted geometry comes from site voxels. Voxel ownership is then fixed for all updates.

B. Correspondence and Bidirectional Correction

Design coordinates enter the site frame through

$$\mathbf{p}_d^{\ell} = A\, \pi_{xy}(M \tilde{\mathbf{p}}_d) + \mathbf{t}, \tag{5}$$

where M handles drawing units and preprocessing, and $(A, \mathbf{t})$ is fixed global registration. Local correction updates effective trajectories while retaining original design identities, faces, and openings; auxiliary envelopes are excluded. For observed wall Γ_o and design trajectory D_j,

$$\alpha_{oj} = \arccos |\mathbf{t}_o^\top \mathbf{t}_j|. \tag{6}$$

Candidates satisfying $d(\Gamma_o, D_j) \leq \tau_d$, $\alpha_{oj} \leq \tau_\alpha$, and projection compatibility form $\Pi(o)$, which is reestimated after updates. Unmatched observations remain; unmatched design becomes prior candidates. Opposite faces are grouped by normal offset and kept separate.

At iteration t, $\mathcal{S}^{(t)} = (\mathcal{O}^{(t)}, \mathcal{D}^{(t)}, \Pi^{(t)}, \mathcal{R}^{(t)})$. *Design-constrained fitting* uses fixed voxels $\mathcal{I}_o$. Principal component

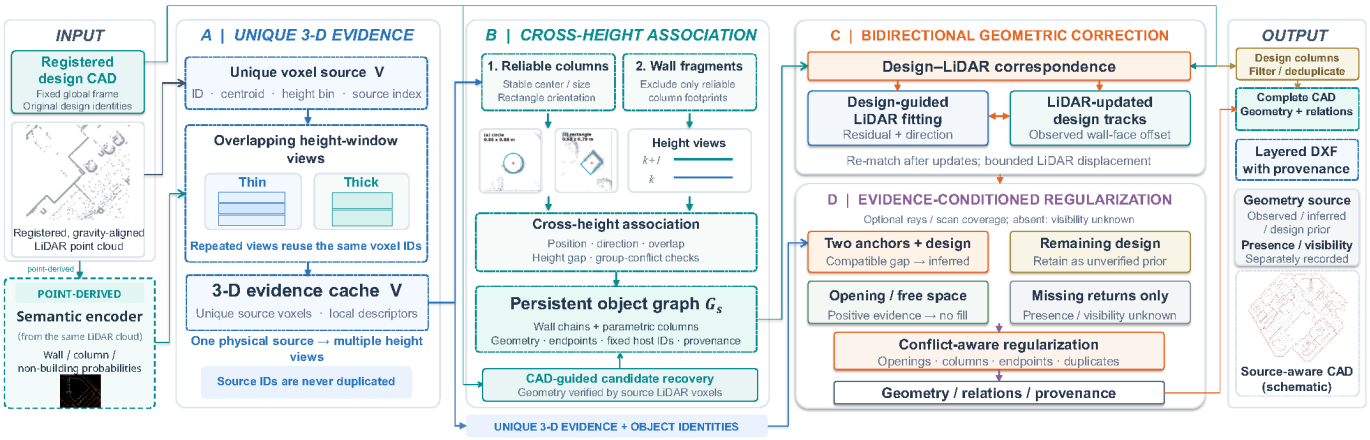


Fig. 2. LiDAR2CAD. A/B: Cross-height site objects and observation-verified recovery. C: Bidirectional correction. D: Anchor-supported completion and unverified design retention.

analysis estimates initial direction $\mathbf{u}_o$ about the planar median. Normal-outlier trimming reestimates center $\mathbf{c}_o$ but not direction. After sign alignment with design,

$$\hat{\mathbf{t}}_o = \frac{(1-\lambda_d)\mathbf{u}_o + \lambda_d \mathbf{t}_{d,o}}{\|(1-\lambda_d)\mathbf{u}_o + \lambda_d \mathbf{t}_{d,o}\|_2}, \tag{7}$$

with $\lambda_d = 0$ when unmatched. Original endpoints projected onto this line form $\hat{\Gamma}_o$. The full host set, not the trimmed subset, evaluates

$$E_o(\Gamma) = \frac{1}{|\mathcal{I}_o|} \sum_{i \in \mathcal{I}_o} d(\pi_{xy}(\mathbf{p}_i), \mathrm{line}(\Gamma)). \tag{8}$$

For original, current, and candidate endpoints $\mathbf{a}^0_{o,k}$, $\mathbf{a}^{(t)}_{o,k}$, and $\hat{\mathbf{a}}_{o,k}$, acceptance requires

$$\begin{aligned} E_o(\hat{\Gamma}_o) + \epsilon_E &< E_o(\Gamma_o^{(t)}), \\ \max_k \|\hat{\mathbf{a}}_{o,k} - \mathbf{a}^0_{o,k}\|_2 &\le \delta_g, \\ \max_k \|\hat{\mathbf{a}}_{o,k} - \mathbf{a}^{(t)}_{o,k}\|_2 &\ge \epsilon_g, \\ \arccos|\hat{\mathbf{t}}_o^\top \mathbf{t}_o^{(t)}| &\le \tau_f. \end{aligned} \tag{9}$$

Here, ϵ_E, δ_g, ϵ_g, and τ_f control residual decrease, displacement from initialization, negligible moves, and angular change. Full-set evaluation prevents apparent improvement from discarding difficult points; coupled-update termination additionally requires stability.

For *observation-constrained design updates*, each observation selects a primary design entity. Reliable multiheight hosts are grouped by normal offset into surfaces $\mathcal{B}_{jf}$. Fitting uses their eligible host-voxel union if at least six unique voxels exist; otherwise, or without evidence, it pools original host endpoints. The planar median and leading principal direction estimate $(\mathbf{c}_{jf}, \mathbf{u}_{jf})$. Endpoint fitting adds no measured support, and neither path uses already regularized walls. Directions beyond the correspondence threshold revert to the original design direction. Host projections define supported interval I_{jf}; only its original endpoints $\mathbf{a}^0_{jf}, \mathbf{b}^0_{jf}$ are updated:

$$\mathbf{a}^{(t+1)}_{jf} = \mathbf{c}_{jf} + \mathbf{u}_{jf}\mathbf{u}_{jf}^\top(\mathbf{a}^0_{jf} - \mathbf{c}_{jf}), \tag{10}$$

and analogously for $\mathbf{b}^{(t+1)}_{jf}$. Overlapping same-surface candidates are deduplicated while retaining links; opposite faces remain distinct. Unsupported intervals retain original geometry and undecided status. After reestimating Π, local associations and connections are checked using host geometry, cross-height evidence, endpoint occupancy, and conflicts. Iteration ends at the limit or after subthreshold geometry change and two rounds with no identity or design-state changes.

C. Evidence-Conditioned Regularization

For a corrected gap I, same-surface anchors $A(I)$ require sufficient length, unique voxels, at least two height bins, vertical coverage, and compatible orientation. With gap tangent $\mathbf{v}_I = (\mathbf{b}-\mathbf{a})/\|\mathbf{b}-\mathbf{a}\|$, host tangents $\mathbf{t}_a, \mathbf{t}_b$, and outward directions $\mathbf{t}_a^{\text{out}}, \mathbf{t}_b^{\text{out}}$,

$$\begin{aligned} |\mathbf{t}_a^\top \mathbf{t}_b| &\ge \cos\tau_A, \\ \max(d^0_\perp, d^{(T)}_\perp) &\le \delta_A, \\ (\mathbf{t}_a^{\text{out}})^\top \mathbf{v}_I &\ge \eta_A, \\ -(\mathbf{t}_b^{\text{out}})^\top \mathbf{v}_I &\ge \eta_A. \end{aligned} \tag{11}$$

Checking offsets before and after correction prevents distinct faces moved closer from becoming one gap.

For direction-compatible original trajectories $\mathcal{D}^0_I$ sharing design identity, continuity is

$$c_D(I) = \frac{\ell\left(I \cap \bigcup_{D^0_j \in \mathcal{D}^0_I} B_{\tau_c}(D^0_j)\right)}{\ell(I)}. \tag{12}$$

Condition $D(I)$ holds if $c_D(I) \ge \eta_D$. Original corridors establish layout continuity; anchors locate inferred geometry, while updated trajectories locate later prior retention.

Let $O(I)$ be a field-confirmed opening or valid free-space crossing, $R_d(I)$ a design remainder from (14), $F_a(I)$ anchor-completion conflicts (opening, column, crossing, or endpoint), and $F_p(I)$ valid free-space conflicts for a remainder. A design opening alone does not confirm on-site openness. Actions follow

$$\mathcal{Q}(I) = \begin{cases} \text{exclude}, & O(I), \\ \text{infer}, & A(I) \wedge D(I) \wedge \neg F_a(I), \\ \text{retain}, & R_d(I) \wedge \neg F_p(I), \\ \text{hold}, & \text{otherwise}. \end{cases} \tag{13}$$

They set, respectively, σ = open with no wall; (σ, ρ) = (present, inferred); (undecided, design); or no added geometry with undecided presence, preserving the independent attributes in (2). A conflict blocks addition but does not prove openness.

Visibility is independent. Explicit coverage confirming no scan gives ν = unscanned; otherwise retained design has ν = unknown. Missing returns alone never provide free space, and failure to detect a conflict does not imply complete coverage. When registered rays $\mathbf{r}(s) = \mathbf{o} + s\mathbf{d}$ exist, only rays crossing the target's valid height before the first return with a margin establish free space; ground returns and foreground occluders do not. Without origins and ray associations, free space is unknown.

Candidates are rejected against openings, reliable columns, free space, nonhost walls, and endpoint competition, then processed by increasing length with stable-ID ties. Endpoint competition applies to new conditional connections, not unverified prior identity. Each free endpoint accepts at most one new connection; accepted candidates constrain subsequent crossings and duplicates. Parallel overlap is clipped and short remainders removed. Anchors must be observed, preventing recursive use of inferred walls. Facility lines never act as structural constraints.

For effective surface D_{jf}^{*}, opening/column exclusions $\mathcal{B}$, and aligned existing geometry $\mathcal{G}_{\parallel}$, the unverified remainder is

$$\widetilde{D}_{jf} = D_{jf}^{*} \setminus \left(\mathcal{B} \cup \bigcup_{g \in \mathcal{G}_{\parallel}(D_{jf}^{*})} B_{\delta_r}(g) \right). \tag{14}$$

Flat longitudinal buffer caps avoid endpoint overtrimming. Connected fragments must satisfy $\ell \geq \ell_{\min}$ and have no valid free-space conflict. Updated supported intervals and unchanged original intervals are each retained once; distinct faces are not merged.

D. Topology and Multilayer CAD

Endpoints and accepted relation attachments index (3). Observed and deduplicated design columns retain provenance; unverified columns remain undecided. Columns constrain conflicts and enter evaluation, but not wall fitting. Adjacency is separate from explicit $\mathcal{R}$; neither line contact nor retained priors creates CONNECT.

Intervals link status, provenance, visibility, source voxels or design entities, and update operation. Design-derived parking outlines, lane centerlines/boundaries, and arrows retain class, source layer, entity identity, and globally transformed coordinates. Local wall corrections update only the structural layer; facilities affect neither structural fitting nor metrics. Both layers share the site frame, but provenance distinguishes measured boundaries from mapped facilities. Displaying a facility beside a corrected wall does not verify their relative placement; facility-to-wall offsets require field validation. Algorithm 1 summarizes reconstruction.

IV. Experiments and Results

Algorithm 1 Bidirectional Correction and Evidence-Conditioned Regularization

Require: $\mathcal{P}$, $\mathcal{D}_0$, fixed global transform, parameters, optional $\mathcal{E}_s$
Ensure: $\mathcal{M}$ with provenance

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Voxelize; associate walls across heights; extract columns; verify design-guided recovery
Fix source ownership and initialize O^(0), Π^(0)
for t = 0, ..., T − 1 do
    Refit observations by (7) subject to residual/displacement tests
    Update supported design intervals by (10); retain unsupported intervals and face identities
    Reestimate correspondences, associations, and local connections
    Stop after the specified two-round stability condition
end for
Generate gaps from observed anchors and continuous design
Apply (13) and geometric/endpoint conflicts; clip duplicates
Assign status, provenance, and visibility; organize geometry and explicit relations; export CAD
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A. Experimental Setup

ICPARK, an underground garage in Chengdu, China, supports controlled comparisons and ablations; facilities A/B/C provide independent tests. The lane-following Ackermann platform in Fig. 3 carries three-dimensional LiDAR, an inertial measurement unit (IMU), wheel odometry, and edge computing. Registered, gravity-aligned maps support postmapping structural evaluation.

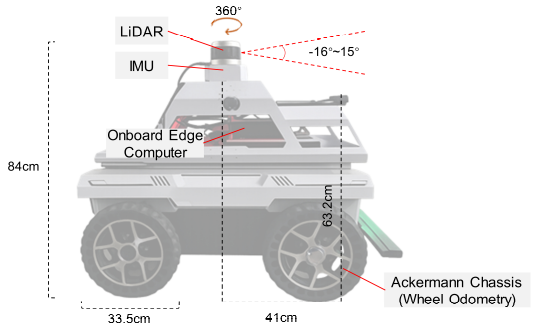


Fig. 3. ICPARK acquisition platform, sensor fields of view, and dimensions. The Ackermann chassis provides wheel odometry.

ICPARK contains 10,742,193 raw points; removing ground, ceiling, and residual slabs leaves 4,720,223 points and 4,548,561 unique voxels. Design CAD supplies walls, columns, and openings. Fixed refined registration is used throughout, excluding envelopes and nonstructural layers. LiDAR-only, overlay, one-way correction, and Full share the extraction front end; common-initialization tests also fix objects and source voxels.

Two development and four evaluation regions each comprise a 20×20 m core plus 5 m context. Common references contain 82 wall segments (266.412 m), 83 column-boundary fragments (148.336 m), 46 CONNECT, 27 OPEN, and four

TABLE I
Shared reconstruction settings. Lengths are in m unless stated otherwise. Ablations change only the specified modules.

Parameter	Value
Voxel grid g / height bin Δz	0.05 / 0.10
Thin window width / step	0.50 / 0.30
Thick window width / step	1.00 / 0.50
Association angle / lateral deviation	10° / 0.035
Association maximum height gap	1.00
Wall probability: extraction / refit / recovery	0.80 / 0.50 / 0.35
Correspondence τ_d / τ_α	0.50 / 12°
Refit λ_d / δ_g / τ_f	0.15 / 0.075 / 6°
Acceptance ϵ_E / ϵ_g	0.1 / 1.0 mm
Minimum fitting voxels / height bins	6 / 2
Maximum correction iterations T	12
Anchor length / voxels / height bins	0.50 / 8 / 2
Anchor vertical coverage	0.45
Anchor τ_A / δ_A / η_A	3° / 0.06 / 0.98
Conditional gap length	0.30–6.00
Design continuity τ_c / η_D	0.50 / 0.98
Opening / column exclusion buffer	0.20 / 0.04
Duplicate buffer δ_r / minimum $\ell_{\min}$	0.04 / 0.15

undecided relations. Initialized from fused geometry, they were manually corrected entity by entity without an independent survey network; undecided relations are not scored. Facilities A–C use separate references.

Regions were preselected by occupancy and spatial distribution, with identical crops. Lower-left map coordinates for dev-01/02 are $(115, 175)$ and $(35, 65)$ m; those for eva-01/02/03/04 are $(155, 5)$, $(275, 130)$, $(210, 215)$, and $(10, 170)$ m. Each axis spans 20 m. Their 0.05 m-grid occupancies are 33.16%, 16.57%, 11.29%, 1.72%, 12.24%, and 2.21%. Table I lists the shared reconstruction settings.

Wall probabilities average three classification heads trained on locally annotated surface voxels, using frozen Sonata/Point Transformer V3 features [25], [26] and local geometry. Window extraction requires wall probability ≥ 0.80 and agreement from at least two heads; refitting and recovery use thresholds of 0.50 and 0.35, respectively, with wall as the most probable class. These experiments evaluate reconstruction conditioned on this semantic front end, rather than held-out semantic classification.

Figs. 4–6 illustrate reconstruction stages; quantitative comparisons use the protocol below.

B. Evaluation Metrics and Baselines

Geometry and completeness. Let $\widehat{\mathcal{G}}$ and $\mathcal{G}$ be predicted and reference curve sets, $B_\tau(\cdot)$ a distance-τ buffer, and ℓ arc length. Precision P_τ, recall R_τ, and F1 are computed separately for walls and column boundaries:

$$P_\tau = \frac{\sum_{\hat{g}} \ell(\hat{g} \cap B_\tau(\mathcal{G}))}{\sum_{\hat{g}} \ell(\hat{g})}, \qquad (15)$$

$$R_\tau = \frac{\sum_{g} \ell(g \cap B_\tau(\widehat{\mathcal{G}}))}{\sum_{g} \ell(g)}, \qquad F_{1,\tau} = \frac{2P_\tau R_\tau}{P_\tau + R_\tau}.$$

The primary tolerance is 0.10 m, with tests at 0.05 and 0.15 m. Entity lengths are summed before aggregate ratios; reference coverage uses the union of prediction buffers. False-positive (FP) and false-negative (FN) lengths lie outside the reference buffer or remain uncovered. Median and 95th-percentile (P95) distances use in-tolerance matches.

Throughout the tables, $\uparrow$ and $\downarrow$ indicate that higher and lower values are better, respectively; neutral parameters, sample sizes, and descriptive counts are left unmarked.

Topology and CAD quality. CONNECT uses one-to-one matching with 0.10 m endpoint/host thresholds and 12° orientation tolerance. OPEN separately counts incorrect explicit connections and complete-geometry crossings. Removing 0.10 m at both ends leaves 23 of 27 OPEN interiors evaluable. This is a reference-line conflict metric, not blocked area or traversability. Duplicate length uses 0.05 m/6° criteria; dangling endpoints and components use 0.10 m adjacency. Adjacency does not replace explicit relation accuracy.

Baselines. The seven external line-extraction baselines described in Section I-B use the same cleaned cloud and evaluation crops to compare complete extraction pipelines.

Shared-front-end LiDAR-only omits design; Overlay unions it with CAD-only; One-way disables only observation-driven design updates and corresponds to *No reverse update*. Non-shared is an overall pipeline comparison, not a single-factor test. Shared-front-end and common-initialization tests isolate bidirectionality.

FLD and EDLines use the same 4,720,223 cleaned points and a 0.10 m full-scene density projection. OpenCV-contrib 4.10.0 default detection parameters are used, followed by a 0.25 m minimum-length filter, without evaluation-region tuning.

C. Structural Reconstruction Performance

Table II reports wall P/R/F1 of 87.95%/84.98%/86.44%, exceeding CAD-only, overlay, and one-way F1 by 33.30/15.03/5.09 points. Against shared-front-end LiDAR-only, recall increases from 55.00% to 84.98% and FN length falls from 119.89 to 40.00 m.

Overlay has 149.20 m FP, versus one-way's 39.75 m and Full's 31.88 m. Full predicts 264.50 m against 266.41 m reference length; median/P95 matched distances are 21.0/78.0 mm.

Table III reports regional results. Wall F1 in eva-04 is 81.95%, versus CAD-only's 36.13%; eva-01/02/03 achieve 97.75%/98.74%/96.50%.

Column-boundary F1 is 93.59%, versus CAD-only's 70.00% and overlay's 81.96%; predicted/reference lengths are 148.21/148.34 m, with 8.23/10.76 m FP/FN.

CONNECT TP/FP/FN=42/3/4 yields P/R/F1 of 93.33%/91.30%/92.31%; the candidate pool matches 45/46 references (97.83% recall). No explicit relation violates OPEN, but complete geometry crosses 1/23 evaluable intervals, confirming that relation decisions and geometry consistency differ.

D. Design–LiDAR Joint Reconstruction Analysis

Common initialization and bidirectional gains. Table IV compares one-way and bidirectional updates with identical observed objects and source voxels. Observed-wall F1 increases from 70.68% at initialization to 74.46% with one-way and

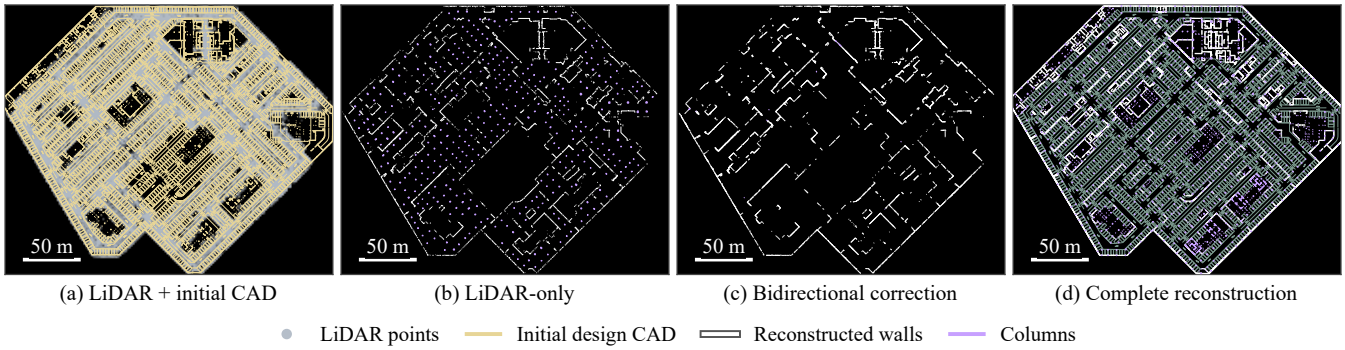


Fig. 4. ICPARK stages at common extent/scale: (a) points and initial design; (b) LiDAR-only; (c) bidirectional correction; (d) complete CAD, including mapped facilities. Metrics concern structures only.

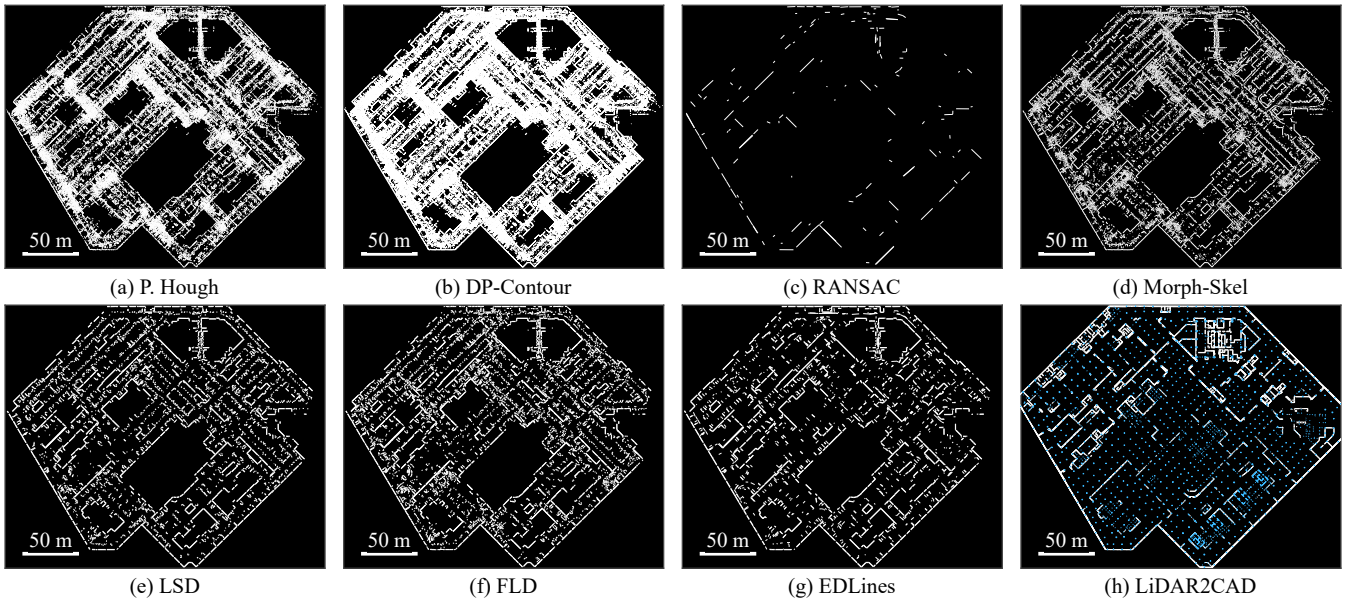


Fig. 5. Global reconstruction at common extent/scale: (a)–(g) seven external baselines; (h) LiDAR2CAD. Baseline names appear above each panel.

78.20% with bidirectional correction, a 3.74-percentage-point advantage for the latter. Including Stage I inferred walls gives 81.82% versus 77.58%, a 4.24-percentage-point difference. Local updates affect 600 design entities. Subsequent regularization raises wall F1 from the Stage I value of 81.82% to 86.44% for the complete output.

Fixed-object scores isolate geometric changes. Expanded Stage I/complete-output scores are not additive module contributions; updated-entity counts measure extent, not accuracy.

E. Ablation Studies

Table V independently disables modules under identical settings. No reverse update stops observation-driven design updates; no design direction sets $\lambda_d = 0$ in (7); no dual anchors still allows unverified prior retention; no prior retention removes only that source; and no regularization conflicts disables opening, column, nonhost-crossing, and endpoint tests.

Without reverse updates or design-direction correction, wall F1 drops 5.09 or 6.55 points and OPEN crossings rise from one to three. Without dual anchors, F1 drops 1.63 points, correct connections fall from 42 to 33, and CONNECT F1 from 92.31% to 80.49%; retained design masks some wall-coverage loss. A retained line can cover a gap geometrically without establishing a supported relation between its hosts. Without unverified priors, wall F1 is 84.42% while connections remain 42/3/4. Thus, observed and inferred geometry account for most of the complete-output F1, while prior retention provides additional coverage without creating accepted connections.

Removing regularization conflict checks increases recall from 84.98% to 89.00%, but reduces precision to 79.00%, increases incorrect connections from three to 16, and raises OPEN crossings from one to seven. Conflict checks trade some recall for higher precision and fewer structural conflicts.

The modules address distinct problems: measured geometry, supported relations, complementary coverage, and structural consistency. Wall F1 alone cannot distinguish them. Association and deduplication lack separate ablations; neither receives an individual accuracy-gain claim.

F. Robustness and Failure Analysis

Prior displacement and deletion. After fixed site initialization, X offsets of 0–0.30 m (0.05 m steps) and 10%/30% entity deletions are independently applied within each core plus context. Deletion removes all entity segments. Thus,

TABLE II
Four-region geometry at $\tau = 0.10$ m. P/R/F1 in %; FP/FN lengths in m. LiDAR-only, Overlay, One-way, and Full share the front end; Non-shared uses a separate pipeline.

(a) Walls

Method	P↑	R↑	F1↑	FP↓	FN↓
P. Hough [11]	5.39	14.00	7.78	636.19	229.12
DP-Contour [13]	3.77	13.54	5.90	976.85	230.34
RANSAC [12]	34.49	1.01	1.97	5.81	263.72
Morph-Skel [14]	10.43	9.35	9.86	188.08	241.51
LSD [15]	14.31	11.57	12.79	176.20	235.58
FLD [16]	12.16	11.81	11.98	198.87	234.94
EDLines [17]	16.32	12.39	14.08	159.98	233.42
CAD-only	53.14	53.13	53.14	120.40	124.87
Non-shared	40.21	53.08	45.76	259.80	125.00
LiDAR-only	84.00	55.00	66.47	28.80	119.89
Overlay	65.85	78.00	71.41	149.20	58.61
One-way	85.00	78.00	81.35	39.75	58.61
Full	**87.95**	**84.98**	**86.44**	**31.88**	**40.00**

(b) Column boundaries

Method	P↑	R↑	F1↑
CAD-only	70.17	69.83	70.00
LiDAR-only	94.00	65.00	76.86
Overlay	80.01	84.00	81.96
Full	**94.45**	**92.75**	**93.59**

TABLE III
Regional results for complete geometry ($\tau = 0.10$ m). Lengths are in m; precision (P), recall (R), and F1 are in %. All denotes scores aggregated by length.

Metric	eva-01	eva-02	eva-03	eva-04	All
Ref. wall length	18.29	19.21	39.57	189.34	**266.41**
Pred. wall length	18.50	19.00	40.00	187.00	**264.50**
Wall P↑	97.50	99.50	96.50	84.00	**87.95**
Wall R↑	98.00	98.00	96.50	80.00	**84.98**
Wall F1↑	97.75	98.74	96.50	81.95	**86.44**
Column P↑	94.00	100.00	95.00	88.00	**94.45**
Column R↑	92.00	99.60	93.00	86.00	**92.75**
Column F1↑	92.99	99.80	93.99	86.99	**93.59**

the test measures postinitialization sensitivity, not end-to-end robustness. The local control (85.44%) differs from cropped full-scene reconstruction (86.44%).

In Fig. 7(a), a 0.10 m X offset reduces aggregate F1 from 85.44% to 83.77%, a decrease of 1.67 percentage points. At 0.30 m, F1 falls to 72.71%, a 12.73-percentage-point decrease. Region eva-04 declines from 80.95% to 66.67% across the tested offsets.

In Fig. 7(b), deleting 10% and 30% of design entities yields aggregate F1 of 83.99% and 80.82%, respectively,

TABLE IV
Stage comparison with common initialization (F1 in %, $\tau = 0.10$ m). Observed walls use identical objects. Stage I additionally includes its inferred walls, but excludes Stage II design remainders. Updated denotes the number of updated design entities.

Stage	Observed wall F_1 ↑	Stage I wall F_1 ↑	Updated
Initialization	70.68	70.68	0
One-way	74.46	77.58	0
Bidirectional	**78.20**	**81.82**	**600**

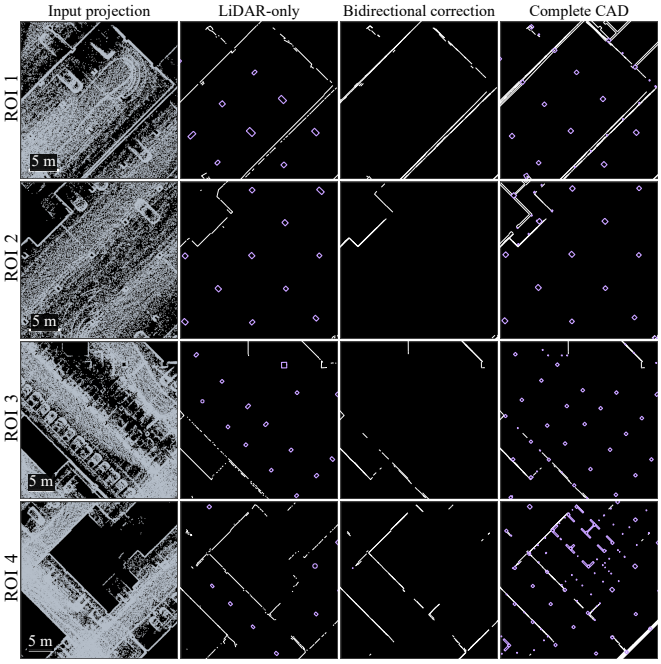


Fig. 6. Local stages with common row extent/scale (bars: 5 m). White/purple: walls/columns. Display ROIs 1–4 are not evaluation regions eva-01–04.

decreasing by 1.45 and 4.62 percentage points from the local unperturbed control. These findings are limited to the tested offset direction, magnitudes, and deletion samples under fixed initialization.

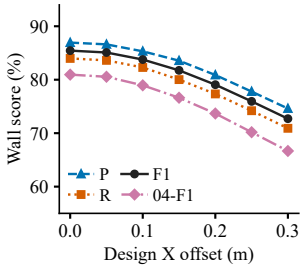
Translation degrades both precision and recall, consistent with misplaced design constraints affecting correspondence and recovered geometry. Deletion primarily reduces recall: at 30% deletion, recall falls from 83.98% to 74.85%, while precision remains 87.84%. This pattern is consistent with loss of complementary design coverage while the fixed observation initialization remains available.

Remaining failures. Fig. 7(c) localizes connection errors to eva-04: candidate coverage is 25/26, final TP/FP/FN is 22/3/4, and OPEN crossings are 1/6. Other regions recover all reference connections without OPEN crossings; eva-02 has no OPEN samples. Aggregate candidate coverage of 45/46 versus final recall of 42/46 leaves one reference uncovered during generation and three not recovered subsequently. Improving candidate coverage alone therefore cannot eliminate missed relations. Counts locate diagnostic stages, but cannot attribute individual errors to repeated surfaces, host endpoints, or conflict arbitration. Eva-04 contains 71.07% of reference wall length, so aggregate accuracy is not uniform local reliability. OPEN excludes width and blocked area; engineering implications require site validation.

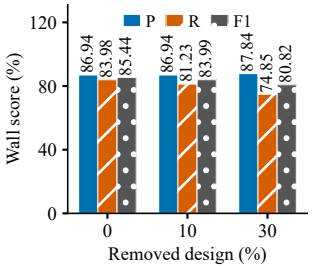
Scoring tolerance. Fig. 7(d) changes scoring tolerance for fixed outputs, separately from input-prior perturbations. Full-method wall F1 is 77.53%/86.44%/92.19% at 0.05/0.10/0.15 m, respectively, exceeding CAD-only, shared-front-end LiDAR-only, overlay, and one-way correction at each tolerance. At 0.05 m, overlay achieves 54.56%, below LiDAR-only's 59.43%, showing that adding insufficiently aligned design lines need not improve geometric accuracy.

TABLE V
Single-factor ablations. P/R/F1 denote wall precision/recall/F1 (%); ΔF_1 is the change from Full in percentage points. OPEN counts complete-geometry crossings out of 23 intervals. CONNECT reports true positives/false positives/false negatives (TP/FP/FN).

Configuration	P↑ (%)	R↑ (%)	F1↑ (%)	ΔF_1 ↑	OPEN↓	CONNECT TP↑/FP↓/FN↓
Full	**87.95**	**84.98**	**86.44**	**0.00**	**1**	**42/3/4**
No reverse update	85.00	78.00	81.35	-5.09	3	37/6/9
No design-direction correction	83.00	77.00	79.89	-6.55	3	35/7/11
No dual-anchor completion	89.00	81.00	84.81	-1.63	1	33/3/13
No unverified prior retention	92.00	78.00	84.42	-2.02	1	42/3/4
No regularization conflicts	79.00	89.00	83.70	-2.74	7	43/16/3



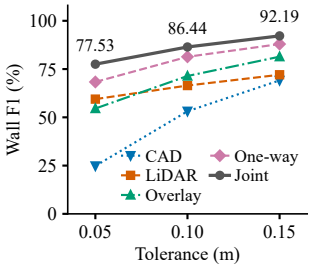
(a) Design translation



(b) Design deletion

ROI	C/N	TP	FP	FN	O
01	9/9	9	0	0	0/4
02	6/6	6	0	0	N/A
03	5/5	5	0	0	0/13
04	25/26	22	3	4	1/6
All	**45/46**	**42**	**3**	**4**	**1/23**

(c) Regional topology



(d) Scoring tolerance

Fig. 7. Prior sensitivity, relation recovery, and scoring tolerance. (a), (b) Aggregate wall precision (P), recall (R), and F1 with fixed initialization, local context, and 0.10 m tolerance; perturbations are independent, and 04-F1 denotes eva-04. (c) Regions of interest (ROI) 01–04 denote eva-01–04; All is the aggregate. C/N: matched CONNECT candidates/reference count; TP/FP/FN: true positives/false positives/false negatives; O: complete-geometry OPEN crossings/evaluable intervals; N/A: no samples. (d) Fixed, cropped full-scene outputs are rescored without rerunning reconstruction; CAD/LiDAR denote CAD-only/LiDAR-only.

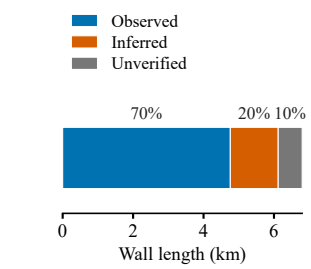
TABLE VI
Correction, regularization, and export costs. Time is in s and memory in GiB. Front-end processing and ray preprocessing are excluded.

Stage / metric	Median / peak↓	Run range↓
Bidirectional	420 s	402–438 s
Regularization	40 s	39–41 s
CAD/provenance export	4 s	4–4 s
Total	464 s	445–483 s
Peak memory	2.20 GiB	2.10–2.20 GiB

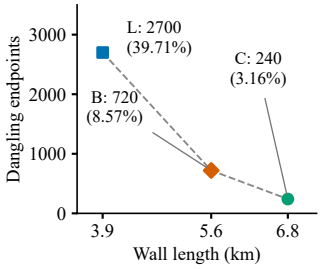
G. Efficiency and CAD Quality

Timing starts after loading initialized objects and evidence, covering correction, regularization, and CAD/provenance export. It excludes simultaneous localization and mapping (SLAM), training, semantic inference, extraction, and ray preprocessing. Three identical-hardware runs report median and range; memory reports peak and range.

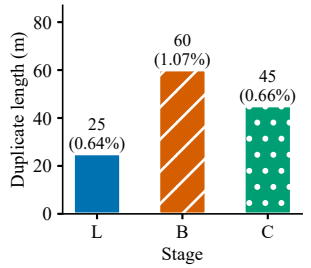
Median total runtime is 464 s, ranging from 445 to 483 s, with peak memory of 2.20 GiB. Correction dominates at 420 s; regularization and export take 40 and 4 s. Stability is reached at iteration six, with maximum geometric change of 0.8 mm and zero identity changes or design updates for two consecutive rounds.



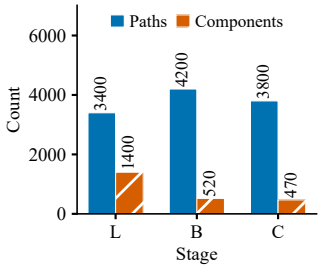
(a) Geometric provenance



(b) Length and dangling ends



(c) Duplicate wall length



(d) Paths and components

Fig. 8. Full-scene provenance and CAD quality. (a) Observed/inferred/unverified design: 4,760/1,360/680 m. (b)–(d) L/B/C: LiDAR-only/bidirectional/complete. Parentheses give dangling-endpoint ratios; dashed lines connect stages. Duplicate length uses 0.05 m/6° criteria; endpoints/components use 0.10 m adjacency.

Fig. 8 assesses full-scene CAD quality, not four-region accuracy. From LiDAR-only through bidirectional correction to complete output, wall length rises from 3,900 to 5,600 and 6,800 m, and dangling endpoints fall from 2,700 to 720 and 240. Regularization reduces duplicate length from 60 to 45 m and components from 520 to 470. Independent faces and openings permit multiple components; these organizational metrics do not replace geometric or relation accuracy.

Provenance-link completeness is 100%: CAD entities trace to observed objects, inference anchors, or design entities. This measures traceability, not field-verification accuracy. Unverified design and inferred intervals without sufficient scan evidence retain uncertainty attributes.

H. Additional Facilities

Table VII summarizes three independent facilities. Wall F1 for A/B/C is 83.95%/81.42%/82.42%, exceeding their CAD-only baselines by 17.95/19.42/18.42 percentage points.

TABLE VII
Joint reconstruction at independent facilities A/B/C. Wall and CONNECT F1 are in %. CONNECT counts true positives/false positives/false negatives (TP/FP/FN); OPEN reports crossings/reference intervals.

Metric	A	B	C
Reference CONNECT	60	80	50
CAD-only wall F1↑	66.00	62.00	64.00
Joint wall F1↑	83.95	81.42	82.42
CONNECT TP↑/FP↓/FN↓	55/4/5	70/6/10	44/4/6
CONNECT F1↑	92.44	89.74	89.80
OPEN crossings↓/reference	1/30	2/40	1/25

CONNECT F1 ranges from 89.74% to 92.44%, with remaining OPEN conflicts of 1/30, 2/40, and 1/25.

Facilities A/B/C have separate references with 60/80/50 CONNECT relations and 30/40/25 OPEN intervals. Scores use each facility's CAD-only baseline and remain separate to expose sample-count and performance differences. These structural comparisons do not establish unchanged semantic preprocessing, transfer without retuning, or field accuracy of mapped facilities.

V. Conclusion

LiDAR2CAD combines bidirectional design–observation correction and evidence-conditioned regularization to reconstruct walls and columns. Mapped facilities complete a semantic CAD map retaining provenance and verification status. ICPARK wall/CONNECT F1 reaches 86.44%/92.31%; controlled comparisons support bidirectionality, and wall F1 exceeds CAD-only at three independent facilities.

Limitations include manually reviewed rather than independently surveyed references, fixed-initialization sensitivity tests, and limited facility coverage without established zero-shot semantic generalization. OPEN measures line conflicts, not passage safety; large design discrepancies still cause missed/incorrect connections and opening crossings. Future work will expand facility coverage, verify unobserved intervals using scan poses and rays, evaluate physical passage space, and measure end-to-end costs including extraction and ray preprocessing.

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